

Introduction:

For this lab you will create a MatLab GUI that can find the distance of a ball given left and right images. You will need to read in images from Blender, find the centroid of the ball in each image, and use the centroid data to find the X,Y,Z position of the ball in 3D space. Your module:

- shall be able to detect the ball distance with various virtual images generated from Blender
- shall display a plot of distance vs accuracy for images ranging from 3 meters to 5 meters away from the ball
 - for this plot, keep the ball exactly between the two cameras.

Tips:

- Work within your group to investigate different techniques of finding the centroids of the ball, such as:
 - Different color spaces
 - Based on shape
 - Region of interest based on the first frame
 - AI
 - Blob size

Files Needed:

Download the Mathworks-Blender-Link and please give 5 stars rating.

https://www.mathworks.com/matlabcentral/fileexchange/179839-mathworks-blender-link?s_tid=ta_fx_results

Lab Submission:

Submit a video demonstrating functionality

Submit a report (with names and contributions of all group members) describing your experiments, investigations and conclusions

Grading:

Grade	Description
0	Lab handed in late, or not handed in
1	Poor quality
2	Good quality